

Deciphering the role of the major histocompatibility complex, the intestinal microbiome and metabolites in the pathogenesis of acute graft-versus-host disease

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ABSTRACT

Allogeneic hematologic stem cell transplantation is a cornerstone in modern hematological treatment, yet its efficacy is compromised by acute Graft-versus-Host Disease. In acute Graft-versus-Host Disease, conditioning regimen induced epithelial damage leads to release of damage and pathogen associated molecular patterns which in turn triggers activation of alloreactive donor T cells, ultimately resulting in destruction of healthy tissue. Advances in major histocompatibility complex typing and preclinical studies using tissue specific major histocompatibility complex deletion have illuminated the contributions of both, hematopoietic and non-hematopoietic cells to acute Graft-versus-Host Disease pathophysiology. Concurrently, high-throughput sequencing techniques have enabled researchers to recognize the significant impact of the intestinal microbiome and newly discovered metabolites in the pathophysiology of acute Graft-versus-Host Disease. In this review, we discuss the implications of major histocompatibility complex expression on hematopoietic and non-hematopoietic cells, the effect on the intestinal microbiome and the metabolic alterations that contribute to acute Graft-versus-Host Disease. By combining these findings, we hope to untangle the complexity of acute Graft-versus-Host Disease, ultimately paving the way for the development of novel and more effective treatment options in patients.

1. Introduction

Allogeneic hematopoietic stem cell transplantation (allo-HCT) is a potentially curative therapy for several benign and malignant hematological diseases as well as for different types of immune deficiencies [1]. In allo-HCT, patients undergo treatment with a cytotoxic conditioning regimen, usually comprising of chemotherapy and/or irradiation. Following this intervention, a mixture of different hematopoietic cells, including CD34⁺ hematological stem cells from a human leukocyte antigen (HLA) matched donor (sibling donor, matched unrelated donor, mismatched unrelated donor or haploidentical donor) are transfused back into the patient. Once these cells are engrafted, the newly generated donor T cells can eliminate any residual malignant cells, a beneficial mechanism known as Graft-versus-Tumor (GVT) effect [2]. However, success of allo-HCT is often curtailed by potential complications such as risk

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of relapse, infections, acute organ toxicities and notably occurrence of acute Graft-versus-Host Disease (aGVHD) which significantly impacts mortality after allo-HCT [3–5]. In aGVHD, epithelial damage induced by the condition regimen triggers the release of damage and pathogen associated molecular patterns (DAMPs and PAMPs respectively). These molecular patterns are consequently recognized and presented by antigen presenting cells (APCs), such as dendritic cells (DCs), to the donor-derived T cells, inciting an alloreactive immune response against healthy tissues [6]. In patients, common organ manifestation include the skin, liver and the intestine [7], however in recent years also non-classical GVHD organs have been reported [8]. Corticosteroids remain the first line of treatment for aGVHD [9], yet a significant proportion of patients becomes refractory, facing an unfavorable prognosis with only a 2 year survival of 5–30 % survival [10–12], in particular in the context of steroid refractory intestinal aGVHD [13]. This underscores an urgent need for the development of novel treatment strategies. In recent years, advances in molecular major histocompatibility complex (MHC)-typing have enabled clinicians to further dissect the role of MHC-II and minor histocompatibility antigen expression on non-hematopoietic cells [14,15]. Furthermore, advances in high-throughput sequencing techniques such as 16s rRNA gene sequencing or shotgun metagenomics sequencing, have grown our understanding of how the intestinal microbiome can influence aGVHD [16–18] and have led to novel strategies that aim at maintaining microbial homeostasis while targeting invading bacteria with antibodies [19]. In this context, the influence of antibiotics on the microbiome and aGVHD has surfaced as a matter of critical significance [20,21]. While antibiotics are important to control invading bacteria, broad spectrum antibiotics were also shown to reduce microbial diversity which resulted in more aggressive aGVHD [22]. Finally, over the last decade an increasing number of publications has demonstrated the fundamental role of intestinal metabolites in molding the local immune response by provoking both pro- and anti-inflammatory signals [23–25].

2. Human leukocyte antigen

Human leukocyte antigens (HLAs), the equivalent of the MHC found in other species, represents a series of genes located on chromosome 6 that encode for cell surface proteins responsible for immune surveillance, modulation and tolerance [26]. HLAs are highly polymorphic and are classified into three groups, namely class I, II and III. Current routine HLA typing in most European countries includes typing of certain class I (HLA-A, HLA-B, HLA-C) and II HLA (HLA-DPB1, HLA-DQB1, HLA-DRB1) antigens. In allo-HCT patients, the degree of donor-recipient HLA matching is associated with both, risk of relapse and aGVHD [27–30]. Fully matched donors are compatible for 8/8 (HLA-A, HLA-B, HLA-C, HLA-DRB1) or 10/10 (also HLA-DQB1) alleles and are preferred over unmatched donors. Whereas the role of class I HLA molecules as well as HLA-DQ and HLA-DR have long been studied, HLA-DP, has emerged as a key player in aGVHD within the last two decades, specifically in unrelated donors [15,27,31,32]. Initial data from retrospective analysis on the HLA-DPB1 phenotype suggested that mismatching of HLA-DP did not affect survival but increased the risk for aGVHD while reducing relapse risk [27,33,34]. However, subsequent analysis challenged this notion by identifying specific haplotypes [35]. To end controversy, a milestone study by Petersdorf et al. investigated the influence of HLA-DPB1 mismatching on aGVHD risk in over 19,000 patients undergoing unrelated donor transplant. The expression level as well as the total number of HLA-DPB1 mismatch of the patient significantly correlated with the risk of aGVHD [36]. Negative regulation of MHC expression by microRNA-146a via targeting JAK/STAT-signaling in dendritic cells was shown to reduce aGVHD [37], which could be also extended to immune checkpoint inhibitor (ICI) induced adverse events [38], which share similarities with aGVHD [39]. Beyond the traditional understanding of HLA disparity, the concept of non-hematopoietic HLA/MHC expression in aGVHD has been gaining traction. In 2011, Koyama et al. published a preclinical study, illustrating that recipient non-hematopoietic cells have the potential to instigate aGVHD at a potency of 100–1000 times higher than that of donor APCs [14]. Besides classical APCs like dendritic cells also monocytes and neutrophils have been shown to contribute to the activation and priming of incoming donor T cells [40,41]. Furthermore, recent investigations have provided valuable insights into how MHC-II upregulation by intestinal epithelial cells is influenced by the microbiota [42]. While reduced MHC expression is desirable in the context of aGVHD, the GVT effect heavily relies on MHC expression on the malignant cells [43,44] and strategies to enhance MHC expression discovered in the mouse model [45] are actively tested in the clinical trial setting (NCT05447663). Although these discoveries are yet to be confirmed in patient studies, they offer intriguing perspective on how the non-hematopoietic cells and the microbiota impact aGVHD pathophysiology and how targeting these might provide future patients benefits.

3. Microbiome

Under physiological conditions, the gut microbiome, housing trillions of diverse microorganisms, maintains intestinal homeostasis and promotes tolerance through interactions with itself but also metabolites, the intestinal epithelium and the immune system [46–51]. Pioneering research on the composition of intestinal bacteria in mice during the 1970s established the value of a diverse bacterial microbiome in mice [52,53] and was subsequently confirmed in patients [54]. The bacterial microbiome in healthy individuals is rich and non-pathogenic, mainly comprising of the phyla *Firmicutes*, *Bacteroidetes*, *Actinobacteria*, *Proteobacteria*, *Fusobacteria* and *Verrucomicrobia*. However, the majority of allo-HCT patients exhibit a limited microbial diversity. Studies have linked the degree of microbiome diversification, measured by the inverse Simpson index, to mortality after allo-HCT [17]. Taur and colleagues reported that the overall 3 year survival for patients with the highest diversity was 67 %, compared to 60 % and 37 % for those of intermediate and low diversity, respectively. Furthermore, a study conducted by Pelled et al. utilizing retrospective 16S rRNA gene sequencing analysis from four centers and 1362 patients corroborated that lower diversity was associated with higher risk of death and dominance by single taxa [55]. These findings align with a study by Jenq et al. that revealed significant microbial changes in aGVHD patients in the *Firmicutes* phylum, specifically a decrease in *Clostridiales* and an increase in *Lactobacillales* [56]. Notably, both *Clostridiales* and *Lactobacillales* are heterogeneous orders consisting of different genus of which some have been shown to influence aGVHD

and will be discussed in the following. In pediatric allo-HCT patients, antibiotic-induced depletion of *Clostridia* spp. correlated with increased incidences of aGVHD [18]. Interestingly, the authors also demonstrated that oral administration of *Clostridia* spp. could alleviate aGVHD symptoms in a mouse model. A recent case report of a 22-year-old male with chronic GVHD supported these findings, suggesting that treatment with *Clostridium butyricum* resolved symptoms of tongue leukoplakia and increased the number of regulatory T cells (Tregs) [57]. Currently, a clinical phase 1 trial assessing the safety of *Clostridium butyricum* is active but not recruiting (NCT03922035). This study analyzes CBM588, that may increase gut bacteria biodiversity, and may prevent recurrent symptoms of gastrointestinal toxicity. Meanwhile, Golob et al. discovered a negative correlation between the abundance of study on *Lachnospiraceae* and aGVHD incidence at neutrophilic recovery [58]. The Rousseau group confirmed these findings and identified two subpopulations, *Blautia* and *Ruminococcaceae* families, which responsible for this effect [59]. However, randomized controlled trials investigating *Blautia* and *Ruminococcaceae* family are still lacking. In contrast, aGVHD patients have also shown an increase in *Lactobacillales* [56]. Among *Lactobacillales*, the genus enterococcus is of specific interest due to its high pathogenicity, potential to acquire vancomycin resistance and association with severe complications like bacteremia. Taur et al. conducted a prospective 16S rRNA gene sequencing analysis and demonstrated a three-fold increase in *Enterococcus* in allo-HCT patients after metronidazole application [60]. Additionally, their finding revealed that an increase in enterococcal dominance led to a nine-fold increase in the risk of vancomycin resistant *Enterococcus* (VRE), significantly impacting patients outcome. Nevertheless, data concerning other *Lactobacillus* species is inconclusive with some studies suggesting potential benefits in aGVHD, while others do not. In a mouse model of aGVHD, combination therapy with tacrolimus and *Lactobacillus acidophilus* showed promising results by increasing Treg/Th17 ratio and thereby ameliorated aGVHD [61]. However, a randomized controlled trial on *Lactobacillus rhamnosus* probiotics in allo-HCT patients was terminated as it did not demonstrate protective effects on the microbiome or aGVHD incidence [62]. Consequently, further research will be required to thoroughly assess the impact of *Lactobacillus* in patients with aGVHD. Apart from microbial disturbances caused by allo-HCT itself,

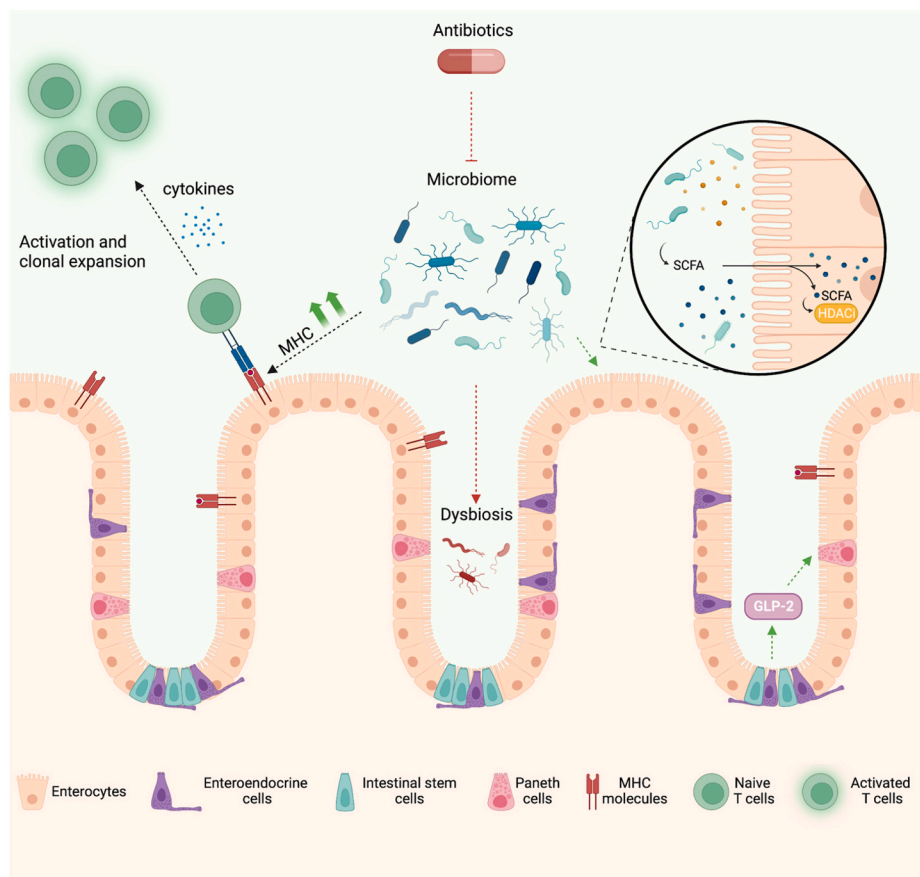


Fig. 1. Interplay between the intestinal microbiome and short chain fatty acids, dysbiosis and MHC-II expression:

Dietary fibers are broken down by the microbiota into SCFAs which influence recipient's epithelial cells by histone deacetylation inhibition, thereby limiting tissue damage induced by conditioning regimen. GLP-2, a peptide hormone secreted by enteroendocrine L-cells, has protective effects on Paneth cells during aGVHD pathophysiology. Antibiotic use drastically reduces microbiome diversity, leading to dysbiosis and allowing translocation of PAMPs, thereby aggregating aGVHD. Additionally, the microbiome can shape MHC-II expression on intestinal epithelial cells, which in turn impacts alloreactive donor T cell activation, proliferation, and end organ damage.

SCFA: short chain fatty acids, GLP-2: glucagon like peptide-2, PAMPs: pathogen associated molecular patterns, aGVHD: acute Graft-versus-Host Disease. MHC: major histocompatibility complex.

pre-, peri- and postinterventional therapy can also affect the intestinal microbiome. In patients undergoing allo-HCT, widespread use of antibiotics is common to prevent or manage complications like neutropenic fever but their usage can disrupt the microbiome and affect patients outcome [63,64] (Fig. 1). Interestingly, type and spectrum of the used antibiotic plays a critical role. A cohort analysis of 857 allo-HCT patients observed that, neither the narrow spectrum antibiotics aztreonam nor cefepime showed an increased GVHD mortality at 5 years. However, broader spectrum antibiotics imipenem-cilastatin and piperacillin-tazobactam were associated with an increase in GVHD related mortality of 21.5 % and 19.8 % respectively when compared to the untreated patients (13,1 % and 11.9 % respectively) [65]. Moreover, timing of antibiotic administering is essential. A retrospective, multicenter study of 621 patients undergoing allo-HCT revealed that early use of broad spectrum antibiotic before allo-HCT was associated with higher transplant related mortality (TRM) (34 %) compared to antibiotic use after allo-HCT (21 %) or no antibiotics (7 %) [20]. The authors attributed this effect to the decreased abundance of commensal *Clostridia* in patients who received antibiotics earlier and called for the development of specific clostridia sparing broad spectrum antibiotics. In this context, fecal microbiome transplant (FMT) therapy has recently emerged as a new promising therapy for steroid refractory aGVHD patients. Initially approved for recurrent pathogenic *Clostridium difficile* infections, pilot studies have deemed FMT safe and feasible in aGVHD patients as well [66]. In addition, different intestinal cell types such as goblet cells, Paneth cells and enteroendocrine cells aid in shaping the microbiome. Paneth cells secrete antimicrobial peptides which influence the local microbiome. Conversely, aGVHD damages Paneth cells and thereby disrupts production of alpha-defensins and intestinal microbial composition [67]. L cells are highly specialized enteroendocrine cells of the intestinal tract which release glucagon like peptide-2 (GLP-2). GLP-2 can induce Paneth cell repair and thereby reduce aGVHD severity in mice [68] (Fig. 1), a concept that is currently tested in a prospective clinical trial (NCT05415410). Other interventions to target the microbiome are summarized in Table 1.

4. Metabolites

In recent years, significant progress in field of metabolomics has been made, shedding light on the intricate interaction among metabolites, the intestinal microbiome, the epithelial lining, and immune cells. Initially explored in preclinical mouse models, mounting evidence now indicates the crucial relevance of these interactions in patients as well [75]. In the following section, we will discuss the impact of fatty acids, aryl hydrocarbon receptor (AhR) ligands/tryptophan derived metabolites and bile acids on aGVHD. Fatty acids are classified into four groups, based on the number of carbon atom they contain: Short chain fatty acids (SCFAs) with 1–6, medium chain fatty acids (MCFAs) with 6–12, long chain fatty acids (LCFAs) with 13–21 and very long chain fatty acids (VLCFAs) with more than 22 carbon atoms. Among these groups, SCFAs have been studied extensively in aGVHD. Butyrate, alongside acetate and propionate the most common SCFAs, has demonstrated to mitigate aGVHD in mice by histone deacetylase inhibition (HDACi) in intestinal epithelial cells and thereby limiting conditioning regimen induced epithelial damage [25,76] (Fig. 1). Additional studies have revealed that butyrate also promotes proliferation of colonic regulatory T cells (Tregs) and stimulate mucus secretion in intestinal epithelial cells [77,78], effectively alleviating aGVHD symptoms. Although butyrate is produced by various intestinal bacteria, pre-clinical models have highlighted the role of *clostridia* (specifically a 17 strain *clostridia* cocktail) in successfully reducing aGVHD severity by increasing butyrate secretion [25]. Patients data on acute and chronic GVHD has confirmed these findings. In 2022, the van den Brink group reported that clostridial abundance and the ratio of strictly to facultative anaerobes were important predictors of overall survival in aGVHD patients [79]. Furthermore, a prospective study analyzing stool samples of 201 patients after allo-HCT revealed that early and prolonged use of broad-spectrum antibiotics diminished the *clostridiace* population and the bacterial butyrate producing enzyme butyryl-coenzyme A (CoA):acetate CoA-transferase (BCoAT) in feces of patients [80]. Decreased BCoAT ratios at GVHD onset correlated not only with gastrointestinal aGVHD severity but also GVHD-associated mortality rate. Consequently, a phase 2 randomized controlled trial is currently underway to investigate the effects of potato-based resistant starch (NCT02763033), hypothesizing that is intervention may increase intestinal butyrate and potentially provide therapeutic benefits [81]. Besides aGVHD, a multicenter trial on the impact of butyrate and propionate in chronic GVHD patients additionally revealed that butyrate and propionate are also protective in chronic GVHD [82]. In aGVHD mouse models, propionate has been additionally been investigated as a ligand of the SCFA sensing G-protein-coupled receptors (GPR). Fujiwara et al. demonstrated that propionate mediated GPR43

Table 1
Interventions to target the microbiome.

Intervention to target the microbiome	Main finding	Clinical or preclinical	Reference
AhR ligands and tryptophan derived metabolites feeding to mice	AhR ligands and tryptophan reduced aGVHD	preclinical	[69–71]
GLP2 substitution in aGvHD mice and grade II-IV steroid refractory aGvHD patients	Expansion of Paneth cells and intestinal stem cells upon GLP2 treatment. GLP2 correlates with NRM and GVHD severity	Preclinical and clinical	[68]
Lipocalin-2 (LCN2) substitution in mice developing aGVHD	Expansion of IL-10 producing macrophages upon LCN2 treatment. LCN2 and aGVHD severity correlates in patients	Preclinical and clinical	[72]
Third-party fecal microbiota transplantation following allo-HCT.	Third-party fecal microbiota transplantation reconstitutes microbiome diversity.	Clinical	[73]
Fecal microbiome transplantation for patients with steroid-resistant, intestinal aGvHD	Increased microbiome diversity	Clinical	[74]

activation lead to ERK phosphorylation with downstream NLRP3 inflammasome activation in intestinal epithelial cells, thereby alleviating GVHD symptoms [83]. In contrast to epithelial cells, NLRP3 activation in myeloid cells of the immune system was shown to contribute to aGVHD severity [84]. However, whether prophylactic propionate administration will provide clinical benefit is still unknown and future clinical trials will be needed to assess this. In addition to therapeutic approaches, fatty acids have also been evaluated as biomarkers for predicting patients outcome after allo-HCT. Gandhi et al. investigated the correlation between plasma SCFA levels and clinical response of 49 steroid depended/refractory patients [85]. Patients with complete or partial response (CR/PR) had higher levels of butyrate and propionate in comparison to non-responder (NR). Additionally, high levels of proinflammatory IL-6 were linked to low plasma concentrations of propionate. Beyond SCFAs, LCFA potential as a biomarker has also been investigated. The stearic/palmitic acid (SA/PA) ratio of 114 patients who underwent allo-HCT was determined by Wu et al. on day 7 after allo-HCT. Notably, patients with higher SA/PA ratio were more likely to develop grade II-IV aGVHD, though no difference in non-relapse mortality or overall survival was noticed. Another crucial group of intestinal metabolites comprises of AhR ligands and metabolites derived from tryptophan. In the years 2009 and 2011, several publications emphasized the significance of kynurenine, a metabolite of tryptophan metabolite produced by the enzyme indolamine 2,3 dioxygenase (IDO), in aGVHD pathophysiology. Initially, preclinical mouse studies demonstrated that several tryptophan metabolites (kynurenine being one of them) could ameliorate aGVHD mortality. Furthermore, it was reported that patients who survived aGVHD patients exhibited lower urinary kynurenine concentrations compared to those that didn't survive or experienced other complications [86]. Notably, lower urinary kynurenine concentrations was correlated with aGVHD severity. The aryl hydrocarbon receptor (AhR) on the other hand is a transcription factor that modulates intestinal immune responses. Interestingly, AhR ligands are mainly produced by tryptophanase found in endogenous bacteria such as Lactobacillus, Proteobacteria or Enterococcus [87]. One example of such metabolite is indole-3-carbinol. A recent preclinical study conducted by Swimm and colleagues recently suggested that intestinal derived indoles can abrogate aGVHD via type I interferon signaling [69]. Indole-3-carbinol (I3C), an already approved nutritional supplement, was able to induce Tregs generation in an in vivo aGVHD mouse model, thereby promoting longer survival [70]. Interestingly, in vitro human Tregs could also be induced by low tryptophan and additional kynurenine [71]. Bile acids represent another diverse group of the intestinal metabolites with its potential therapeutic application currently being examined in aGVHD. Ursodeoxycholic acid (UDCA), a secondary bile acid, has a long history of usage in treating liver conditions such as gallstones, primary biliary cirrhosis and primary sclerosing cholangitis [88]. In prevention of hepatic aGVHD, UDCA significantly reduced serum bilirubin and aminotransferase levels and was able to significantly lower incidence of grade III-IV aGVHD [89]. A 10 year follow-up revealed notable differences in survival (48 % versus 38 %) and non-relapse mortality (28 % versus 41 %), favoring the UDCA-treated group [90]. Moreover, a recent metabolomic analysis comparing HLA-related recipient and donor pairs revealed that at the onset of aGVHD, significant fluctuations within several bile acids could be detected compared to controls without aGVHD [75]. This reports fits in well with a recently published study by the Apostolova group that demonstrated the regulation of intestinal antigen presentation by the bile acid tauroursodeoxycholic acid, leading to alleviation of aGVHD [91] (Fig. 1). Interventions to modify the metabolism are summarized in Table 2. Based on these promising findings, prospective clinical trials investigating the potential of bile acid supplementation are needed to clarify its role in aGVHD prevention.

5. Summary and outlook

Over the past decade, significant progress in our understanding of aGVHD pathophysiology has been made. Notable breakthroughs include new insights into the role of different MHC-subtypes on both hematological and non-hematological cells, recognizing the significance of the intestinal microbiome and the identification of novel metabolites. In forthcoming clinical trials, research might investigate the role certain MHC mismatches on different pro-/anti-inflammatory cells types in aGVHD patients which may provide further risk stratification. Furthermore, the importance of a diverse intestinal microbiome in safeguarding against aGVHD has become evident. While our current understanding mainly revolves around bacteria, other microorganisms such as viruses and fungi may also play an important role in the future. This knowledge will prove particularly valuable for aGVHD patients receiving antibiotics or steroid refractory patients undergoing FMT, a yet rather experimental therapy. In addition to the microbiome, various intestinal metabolites such as fatty acids, AhR ligands and bile acids have deepened our understanding in aGVHD pathophysiology. Looking ahead, clinical perspective will focus on identifying more suitable metabolic candidates that could ameliorate or even protect from aGVHD. However future studies will need to address the challenge of safely and effectively administering the chosen metabolite to their site of action.

Table 2
Interventions to modify the metabolism.

Intervention	Main finding	Clinical or preclinical	Reference
Resistant potato starch (RPS) daily from day –7 to day 100 given to 10 patients undergoing allo-HCT	fecal butyrate levels were significantly higher when participants were on RPS than when they were not	Clinical	[81]
Bicarbonate (NaBi) substitution to patients with AML relapse after allo-HCT	NaBi treatment of post-allo-HCT patients with relapsed AML improved metabolic fitness and interferon-γ production in T cells.	Clinical	[92]

Practice points

Future studies should investigate interplay between HLA expression on intestinal cells and the microbiome for the risks of intestinal GvHD in different matching contexts.

The use of metabolic modifiers against GVHD or to enhance GVL effects (e.g. potato starch to induce butyrate to mitigate aGvHD; NaBi to reduce lactic acid induced T cell inhibition) needs to be investigated in prospective randomized clinical trials.

Transfer of microbiota and the resulting metabolic alterations need to be investigated in prospective clinical trials for their ability to reduce aGvHD.

Research agenda

Looking ahead, clinical perspective will focus on identifying more suitable metabolic candidates that could ameliorate or even protect from aGVHD.

Future studies will need to address the challenge of safely and effectively administering the chosen metabolite to their site of action.

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Competing interests

VW has no competing interests to declare. RZ received honoraria from Incyte, Novartis, Sanofi, Medac and Mallinckrodt all outside of the work reported here.

CRediT authorship contribution statement

Valentin Wenger: Writing – review & editing, Writing – original draft, Visualization, Resources, Investigation, Conceptualization. **Robert Zeiser:** Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Investigation, Formal analysis, Conceptualization.

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